

LENDI PORTAL (v2.2)

Comprehensive System Architecture, Database Schema & API Documentation
Lendi Institute of Engineering and Technology · Generated July 2026

1. Executive System Overview

The **Lendi Portal** is a full-stack web application developed for the Lendi Institute of Engineering & Technology. It provides a digitized, multi-tier outpass approval pipeline, automated student attendance monitoring, role-based administration, and real-time gate security QR code verification. The codebase has been refactored into decoupled **Frontend** (Next.js 16) and **Backend** (Node.js/Express) modules.

Module	Technology Stack	Description
Frontend Client	Next.js 16 (App Router), React 19	Responsive dark glassmorphic UI with role-tailored dashboards.
Backend Server	Node.js, Express.js	RESTful API server running on port 5000 with CORS & JWT middleware.
Database Engine	SQLite (better-sqlite3)	High-performance file-based storage with WAL mode & auto-seeding.
Security & Auth	JWT, bcryptjs	Stateless token verification & salted password hashing.
Verification	qrcode package	Dynamic QR pass generation & Gate Staff exit/entry scanning.

2. Database Schema & Data Models (`backend/lib/db.js`)

The backend utilizes SQLite managed by better-sqlite3 with foreign key enforcement and Write-Ahead Logging (WAL) enabled. The database schema automatically initializes the required tables upon server launch if missing.

Table Name	Primary Fields	Purpose & Foreign Keys
users	id, email, password, name, role, department, created_at	System accounts & roles (student, class_teacher, hod, principal, gate_staff, admin).
students	id, user_id, roll_no, year, semester, section	Student metadata linked to users.id (ON DELETE CASCADE).
outpasses	id, user_id, student_id, reason, destination, from_date, to_date, status, teacher_status, hod_status, principal_status, exit_time, entry_time	Multi-stage outpass request details & timestamp tracking.
notifications	id, user_id, title, message, type, outpass_id, read, created_at	Real-time alert records linked to users.id.
subjects	id, code, name, faculty_name, department, year, semester	Academic course catalogue for student attendance.
attendance	id, student_id, subject_id, date, status	Class attendance logs (present/absent) linked to students.id.

Demo Access Accounts (Auto-seeded)

Role	Email Address	Password	Department
Student	rahul.kumar@lendi.edu.in	password123	CSE
Class Teacher	teacher.cse@lendi.edu.in	password123	CSE
HOD	hod.cse@lendi.edu.in	password123	CSE
Principal	principal@lendi.edu.in	password123	All Depts
Gate Security	gate.security@lendi.edu.in	password123	Security
System Admin	admin@lendi.edu.in	admin123	System

3. Backend RESTful API Reference (`backend/routes/`)

All backend endpoints run under the /api prefix on port 5000. JWT token authentication is enforced via the Authorization: Bearer <token> header.

Endpoint	Method	Auth / Role	Function & Description
/api/auth/login	POST	Public	Authenticates email & password, signs and returns 7-day JWT token.
/api/auth/register	POST	Public (@lendi.edu.in)	Registers new student account & creates student metadata record.
/api/auth/me	GET	JWT (Any Role)	Validates active token and returns current user/student profile.
/api/outpass	GET	JWT (User/Student)	Fetches list of submitted outpasses for the logged-in user.
/api/outpass	POST	JWT (Student)	Creates outpass request & notifies class teacher.
/api/outpass/:id	GET	JWT (Any Role)	Retrieves detailed view of a specific outpass request.
/api/outpass/:id	PATCH	Teacher / HOD / Principal	Approves/rejects outpass at current tier and advances state.
/api/teacher	GET	Teacher / HOD / Principal	Fetches department outpasses filtered by approval status.
/api/hod	GET	HOD / Principal	Returns department student attendance metrics & outpass statistics.
/api/attendance	GET	JWT (Student)	Computes subject-wise attendance percentages & logs.
/api/notifications	GET, PATCH	JWT (Any Role)	Retrieves notifications list or marks notifications as read.
/api/staff/verify	POST	Gate Staff / Admin	Scans approved QR code outpass and logs exit/entry timestamps.
/api/admin/users	GET, POST, DELETE	Admin Only	Manages users, account creation, and user deletion.

4. Multi-Tier Outpass State Machine

Outpass requests follow a strict sequential approval state pipeline. Rejection at any stage immediately terminates the workflow.

Stage	Status Key	Responsible Role	Next Transition State
1. Submission	pending_teacher	Student	Teacher approves → pending_hod
2. HOD Review	pending_hod	Head of Department	HOD approves → pending_principal
3. Principal Approval	pending_principal	Principal	Principal approves → approved
4. Final Approval	approved	Student	Student views & downloads scannable QR pass.
5. Gate Verification	approved	Gate Security Staff	Logs exit_time and entry_time.

5. Frontend Architecture & Routing (`frontend/`)

The frontend is built with Next.js 16 (App Router). During development and production, Next.js proxy rewrites in frontend/next.config.mjs automatically forward all client API requests to the Express backend on port 5000:

```
rewrites() => [{ source: '/api/:path*', destination: 'http://localhost:5000/api/:path*' }]
```

Route Path	View Component	Description & Controls
/	Landing Page	College portal landing page with features overview.
/login	LoginPage (Suspense)	Interactive sign-in with quick role-based demo buttons.
/register	RegisterPage	Student sign-up form restricted to @lendi.edu.in emails.
/dashboard	StudentDashboard	Student hub for active outpasses & attendance summary.
/outpass	StudentOutpassPage	Form to request outpass & view generated QR gate passes.
/teacher/dashboard	TeacherDashboard	Class Teacher queue for reviewing department requests.
/hod/dashboard	HodDashboard	HOD overview for department outpasses & student records.
/principal/dashboard	PrincipalDashboard	Executive overview of outpasses & student attendance.
/staff/outpasses	StaffVerificationPage	Gate Security scanner interface for exit/entry logging.
/admin/users	AdminUsersPage	System administration panel for user account management.